Multiscale Alternately Updated Clique Network for Hyperspectral Image Classification

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***Abstract*— Recently, deep learning has drawn significant attention in hyperspectral image (HSI) classification. With the growth of network depth and feature integration, deep learning demands abundant labeled samples to optimize many parameters. Unfortunately, most hyperspectral data are unlabeled and the available labeled samples are extremely limited. How to obtain richer features under limited training samples is a challenge for HSI classification. To tackle this issue, a new supervised multiscale alternately updated clique network (MSCN) is proposed for HSI classification to fully employ HSI features in different scales. Based on the Clique Block, we design the multiscale alternately updated clique block (MSCB) that applies convolution kernels of various sizes to adaptively exploit the multiscale HSI information and merge them within the block. Meanwhile, the recurrent feedback architecture is introduced to reuse high-level visual information and network parameters. The proposed MSCN includes two MSCBs to capture the multiscale spectral and spatial information in turn. The MSCN improves the information flow and the efficiency of parameter tuning through the feedback mechanism and the cross-utilization of multiscale feature. It not only obtains more abstract HSI information, but also reduces the network depth and the number of parameters, thereby improving the classification accuracy under limited samples. To certify the validity of the proposed MSCN, experiments are conducted on three real HSI datasets and compared with multiple state-of-the-art deep learning-based approaches. The experimental results demonstrate that the presented multiscale network achieves superior performance, especially in the case of a small number of training samples.**

***Index Terms*— CliqueNet, deep learning, hyperspectral image (HSI) classification, multiscale.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are composed of hundreds of narrow and continuous spectra, which include the vast amounts of spectral and spatial information for an observed scene. Due to wealth of information in spectral domain, HSIs have found practical utilizations in a wide range of fields, such as smart city [1], agriculture [2], surveillance [3], and so on [4]. The analysis and processing of hyperspectral data have also received great attention, including spectral unmixing [5], super-resolution [6], [7], classification [8]– [10], target detection [11], [12], and so on [13]– [15]. HSI classification, as a core step, assigns a category label to each pixel on the basis of its characteristics. However, such high-dimensional spectral information creates the curse of dimensionality [9]. Meanwhile, it is difficult to obtain labeled samples of HSIs, and the number of available labeled samples is limited [16], which brings about the Hughes phenomenon of supervised classification. Therefore, how to achieve accurate classification under limited samples for HSIs is still a challenge.

Recently, HSI classification approaches based on neural network have also become a research hot spot, as deep learning achieved outstanding performance in other fields, such as natural language processing [17] and target detection [18]. Unlike traditional classification methods that adopt provided features or structured priors to classify [19]– [21], deep learning methods can automatically acquire more robust and abstract information from an original image, thereby obtaining more competitive classification accuracies than traditional methods. In the early stage of HSI classification based on deep learning, neural networks, as feature extraction tools, are used to transform the original image into high-level abstract features. For instance, the stacked autoencoder (SAE) [22] and deep brief network (DBN) [23] models extract deep features in an unsupervised manner and obtain competitive results for tradition classifiers. However, SAE and DBN require vectors as model inputs, which cause spatial information to not be fully utilized.

Convolutional neural network (CNN), which demonstrates excellent performance and uses extensive parameter-sharing to tackle the curse of dimensionality [24], has recently drawn increasing attention in HSI classification. In [25], 2-D CNNs are exploited to acquire spectral and spatial features and bring high-level information into the multilayer perceptron (MLP)

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to implement classification. With the development of deep learning techniques, CNN can be used to solve the HSI classification problem in end-to-end ways without the need for traditional classifiers. Hu *et al.* [26] proposed a five-layer CNN network for HSI classification with spectral information. This architecture does not consider spatial context. It is significant to utilize the structural context information of the spatial dimension due to the local consistency of HSIs [27]. There are roughly two strategies in the existing CNN-based method to achieve spectral–spatial classification: 1) design two branch networks to capture spatial and spectral information separately and 2) acquire joint spectral–spatial features in the same network. For instance, 1-D-CNN and 2-D-CNN are utilized to automatically obtain hierarchical spectral features and abstract spatial features in [28]. Similarly, double-branch multiattention mechanism network (DBMA) [29] and double-branch dual-attention mechanism network (DBDA) [30] are designed with dual branch networks to process spectral and spatial information separately. These methods all connect the spectral subnetwork and the spatial subnetwork in parallel, and spectral–spatial features are fused before being sent to the classifier. Chen *et al.* [31] suggested a new 3-D-CNN model, which simultaneously captures joint spectral–spatial features and obtains better results than 1-D-CNN and 2-D-CNN. 3-D-CNN does not require any preprocessing of HSI and extracts joint spectral–spatial features without destroying the data structure.

Generally, deeper networks are able to capture finer features, which are accompanied by a larger quantity of parameters to be optimized. Only when the sufficiency of labeled training samples is guaranteed, promising results can be provided by CNN-based methods [32]. However, manual annotation of remote sensing data requires a great deal of manpower and financial resources. Only the finite labeled samples are obtained for training in practical situations [33]. The imbalance between abundant parameters and limited availability of training samples may lead to poor classification performance [34]. The appearance of residual networks (ResNets) [35] and dense convolutional networks (DenseNets) [36] makes it possible to optimize deeper networks. The ResNet establishes the identity mapping between input and output for residual learning [37]. This architecture can help alleviate the vanishing gradient problem. Zhong *et al.* [38] introduced ResNet to their spectral–spatial residual network (SSRN), which learns discriminative features through concatenated spectral and spatial residual blocks. Although ResNet is very efficient, not every layer of such a deep network is effective. This redundancy of network layers and feature maps has caused the reduction of parameter efficiency and the growth in computational costs [36]. The input of each layer in DenseNet consists of the output of all previous layers, which ensues direct access to the gradient in each layer, speeds up information flow, and reduces redundancy. Wang *et al.* [39] designed the fast dense spectral–spatial convolution (FDSSC) approach under the inspiration of DenseNet. The spatial and spectral modules of SSRN and FDSSC are connected in series. They have obtained joint spectral–spatial features without other branches and delivered relatively robust classification performance using small numbers of training samples.

In addition to residuals and dense connections, the use of multiscale features can also increase the richness and diversity of features and obtain more robust results [40]. In most methods, the spatial window has a fixed size, while in real cases, the regions of different land covers often have inconsistent size. Thus, more abundant local context information is provided by means of multiscale features. Lee and Kwon [41] proposed a deeper and wider CNN with a multiscale filter bank and residual learning to explore richer local spectral–spatial multiscale information. In [42], the hyperspectral pyramidal ResNet model was proposed, which gradually increases dimensions of feature maps and groups the pyramidal bottleneck residual blocks to improve the straightforward residual model formulation. Gong *et al.* [43] developed a novel multiscale framework with incorporating multiscale filter banks and diversified metric trained as a fully connected layer to capture multiscale and discriminative features for HSI classification. However, these studies perform multiscale feature fusion at the end of each unit or at the end of each network. Information between different scales lacks interactions. Consequently, the correlation of information at different scales is relatively weak. Gao *et al.* [44] designed a multiscale residual network with the mixed depthwise separable convolution, which mixes up multiple kernel sizes in a single depthwise convolution operation and explores spatial features at different scales. This idea can extract multiscale information from each feature map while reducing the number of parameters to be tuned. In [37], a multiscale dense network (MSDNet) was proposed to implement feature extraction at fine and coarse levels. MSDNet integrates information of different levels and different scales, but it also brings a large amount of network parameters and longer training time. In [45], a new multiscale residual block (MSRB) was proposed for image super-resolution to enhance the correlation and utilization of multiscale features. Due to the flow of information between different scales, MSRB can adaptively acquire local multiscale features and realize information combination at different scales.

To keep the balance between the large number of network weights and the small number of available training samples, we focus on designing a more efficient deep learning model, which can alleviate the demand for computing resources while extracting more complex and high-level features. To this end, a multiscale cross-fusion deep learning framework is proposed for HSI classification. Specifically, the multiscale framework fully explores the relationship between the information at different scales by fusing multiscale features in each layer within the multiscale block. The multiscale dense block (MSDB) integrates dense connection and the cross-fusion between different scales to propagate the different scale-level information in a bottom-top fashion and provide more complementary details of scale level. Inspired by CliqueNet [46], we further improve the MSDB by the recurrent feedback mechanism to design the multiscale alternately updated clique block (MSCB). On the one hand, local context from different scales is adaptively transformed into higher level information to strengthen the

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feature representation ability. On the other hand, higher level semantic information is fed into previous layers to modulate the extraction of lower level features directly.

The contributions of this article can be summarized as three aspects.

1) We propose a novel multiscale cross-fusion deep learning framework for HSI classification. Multiscale spatial–spectral information is fully explored by the information interaction between different scales. The cross-scale dense connections of the MSDB are designed to implement more efficient feature extraction and further improve the classification accuracy. It can perform well in the case of small number of training samples.

2) Taking into consideration that many methods increase the classification accuracy at the cost of higher model complexity which lead to greater demand for computing resources, we propose the new multiscale alternately updated clique network (MSCN) to improve the parameter efficiency. The proposed method can implement more feature refinement operations without excessive parameters. The calculation cost is controllable and the ability of model representation is further strengthened.

3) Extensive experiments demonstrate that the proposed MSCN achieves competitive classification performance on several HSI datasets. Especially, compared with two multiscale models, the proposed MSCN performs more robust in the case of limited samples. Comprehensive ablation experiments and parameter analysis verify the effectiveness of each module of the proposed method.

The remainder of this article is organized as follows. Section II briefly reviews the classic network architecture: residual block, dense block, and clique block. Section III describes three multiscale blocks and the framework of the proposed multiscale networks. The details and the discussion of the experimental results are provided in Section IV. Finally, Section V concludes this article.

II. Classic CNN Architectures

A. Residual Block

In residual blocks, a shortcut structure connecting the input and the output is created to address the degradation problem [35]. Assuming the input x_0 of each block, the block comprises L layers. A nonlinear transformation $H_l(\cdot)$, consisting of Batch Normalization (BN), rectified linear units (ReLUs), and Convolution (Conv), is implemented in each layer. We denote the output of the lth layers as X_l. The output of the $(l-1)^{\text{th}}$ layer is regarded as the input of the lth layer, which can be formulated as

$$X_l = H_l(X_{l-1}). \tag{1}$$

Residual blocks add a skip connection that uses an identity function to bypass nonlinear transformations. The output of the residual block can be expressed as

$$X_L = H_L(X_{L-1}) + X_{L-1}. \tag{2}$$

Fig. 1(a) shows the structure of residual block with two layers. Residual architectures through summation deliver the gradient from a bottom layer to a top layer directly, but may obstruct information flow.

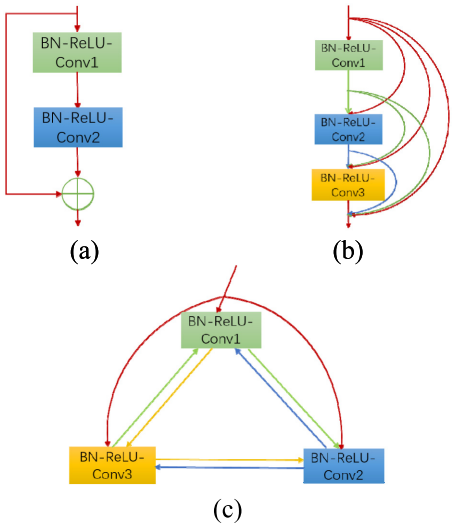


Fig. 1. Classic CNN architectures. (a) Residual block with two layers. (b) Dense block with three layers. (c) Clique block with three layers.

B. Dense Block

In order to further improve the information and gradient flow, the dense connectivity pattern was proposed [36]. Fig. 1(b) illustrates the structure of three-layer dense block, in which each layer has direct connections to all bottom layers. Consequently, the input of lth layer is the concatenation of the outputs of all preceding layers, i.e., $[X_0, X_1, \ldots, X_{l-1}]$, where X_0 denotes the input of dense blocks, $X_i (i > 0)$ is the output of the layer, and [.] refers to the concatenation of feature maps. The dense structure is formulated as

$$X_l = H_l([X_0, X_1, \ldots, X_{l-1}]). \tag{3}$$

With L layers of the dense block, there are $L(L+1)/2$ connections. Dense blocks reduce the interdependence between different layers by reusing feature maps, and the final output consists of the information of all layers to improve the robustness of the network. However, as the increase of network layers, the dense connection paths increase linearly, resulting in a significant increase of parameters [48].

C. Clique Block

Yang *et al.* [46] proposed the clique block in which any layers remain bidirectional connections. Fig. 1(c) shows the layout of three-layer clique block. Compared with dense blocks, clique blocks make each layer not only the input of all later layers but also the input of all earlier layers. The layers of the clique block are constructed in the form of a clique and updated alternately. In this article, we only discuss the case where the number of loops is equal to 2. The update process of a three-layer clique block can be unfolded into two stages and illustrated in Fig. 2. Specifically, the input of clique blocks and feature maps of all preceding layers are connected to update the following layer in stage 1. This process is the same as

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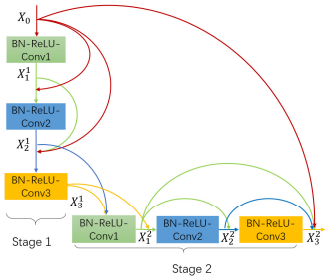


Fig. 2. Propagation of a clique block with three layers.

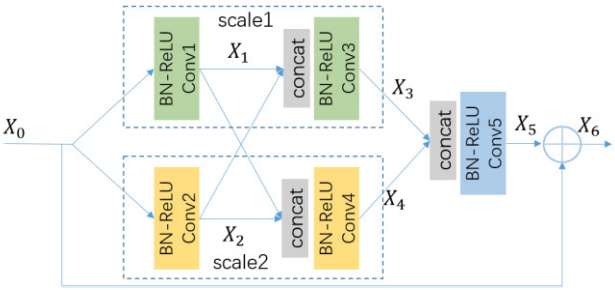


Fig. 3. Structure of MSRB.

dense blocks. In stage 2, feature maps of other layers are conjoined to reupdate the current layer. From stage 2, the block starts updating alternately. Accordingly, the lth ($l \geq 1$) layer of an L-layer clique block in the kth ($k = 1, 2$) loop can be defined as

$$X_l^1 = \left[W_{0l} * g(X_0), ||_{0<i<l} W_{il} * g\left(X_i^1\right)\right] \quad (4)$$

$$X_l^2 = \left[||_{0<i<l} W_{il} * g\left(X_i^2\right), ||_{l<i\leq L} W_{il} * g\left(X_j^1\right)\right] \quad (5)$$

where $g(\cdot)$ denotes the functions of BN and ReLU, $*$ is the convolution operation with W, and $||$ and $[\cdot]$ mean the concatenation of feature maps. In the second stage, the parameter W is reused and the information is looped through the feedback connection. All layers of the clique block are densely connected, whether forward or feedback. This architecture achieves the maximization of information flow and repeated refinement of feature maps.

III. Proposed Framework of Multiscale Networks

A. Multiscale Residual Block

Spectral and spatial information of different scales provide discriminative features to improve HSI classification. It is insufficient to take full advantage of multiscale information to perform multiscale feature concatenation only at the bottom of the unit or the bottom of the network. To address this problem, we introduce MSRBs into HSI classification to improve information flow and increase feature connectivity of different scales in the network. A graphical description of MSRB is displayed in Fig. 3.

MSRB consists of two parts: multiscale feature fusion and local residual learning. MSRB is composed of two bypasses with convolutional kernels of different sizes. Convolution kernels of the same bypass are the same size and belong to the same scale. Through connections between bypasses,

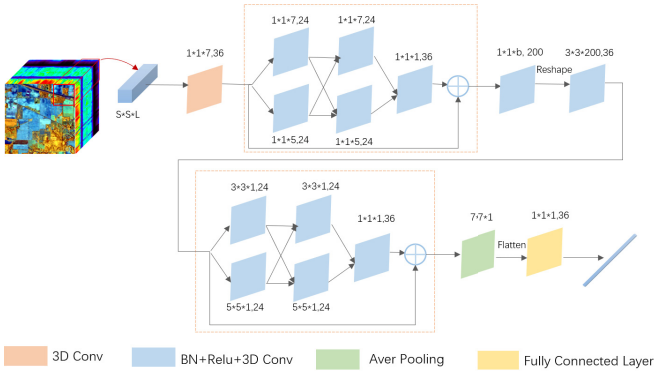


Fig. 4. Overview of spectral–spatial multiscale residual network.

the information between different bypasses is shared with each other. The operations can be formulated as

$$X_1 = W_{01} * g(X_0) \quad (6)$$

$$X_2 = W_{02} * g(X_0) \quad (7)$$

$$X_3 = [W_{13}, W_{23}] * g([X_1, X_2]) \quad (8)$$

$$X_4 = [W_{14}, W_{24}] * g([X_1, X_2]) \quad (9)$$

$$X_5 = [W_{35}, W_{45}] * g([X_3, X_4]) \quad (10)$$

where W_{ij} denotes the parameters of convolution operation, and i and j represent the subscript of input and output feature maps, respectively. In the second layer, feature maps of both input and output are doubled. To enhance calculation efficiency, the bottleneck layer (Conv5) is added to the end of MSRB to reduce channels of the output. For the purpose of improvement of the network performance, MSRB exploits the local residual learning. It can be described as

$$X_6 = X_0 + X_5 \quad (11)$$

where X_0 and X_6 indicate the input and output of MSRB, respectively.

We introduce 3-D-MSRB and propose an end-to-end spectral–spatial multiscale residual network (MSRN) for HSI classification. Fig. 4 shows a graphical representation of the proposed network. For an HSI image with the size H × W × L, the MSRN model selects neighboring pixel patches of S × S × L from the original image as the input, in which H, W, and L refer to the height, width, and spectral dimension of HSI, respectively, and S denotes the spatial size of neighboring blocks used to represent the original information of a center pixel. MSRN is divided into two MSRBs to learn deeper and richer spectral–spatial features in turn. The first 3-D convolution layer with 1 × 1 × 7 convolution kernel processes the neighboring block to obtain initial features as the input of spectral MSRB and reduces spectral dimension. The spectral MSRB, which has two scales of 1 × 1 × 5 and 1 × 1 × 7, extracts and refines high and adaptive spectral features. The bottleneck layer keeps the output size consistent with the input of MSRB. The features after the residual summation are input to two consecutive 3-D convolutional layers, with feature reshaping operation to reduce the spatial size, the number of channels, and the dimensionality. The first transition layer consists of 1 × 1 × b convolution kernels, where b is equal to

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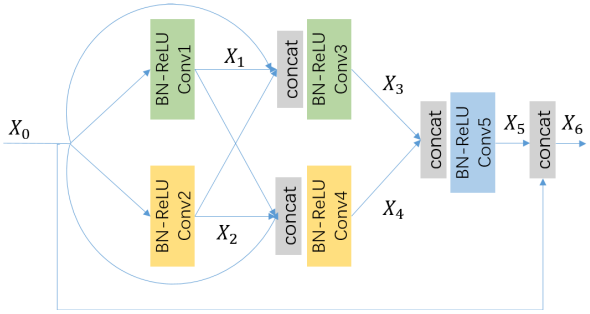


Fig. 5. Structure of MSDB.

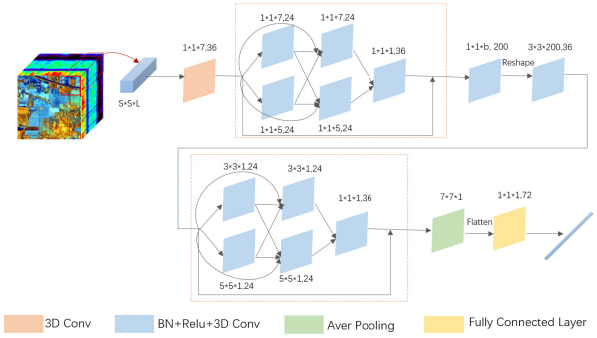


Fig. 6. Overview of spectral–spatial MSDN.

(L-6)/2. New features extracted are sent to the spatial MSRB. Similar to the spectral MSRB, the spatial MSRB adaptively explores multiscale spatial features with $3 \times 3 \times 1$ and $5 \times 5 \times 1$ convolution layers. After a 3-D average pooling layer, the output is flattened into a 1-D feature. Finally, a vector $\hat{y}$ is acquired after the fully connected layer, where $\hat{y} \in \mathbb{R}^{1\times C}$ and C represents the total of categories.

Being good at multiclassification problems, the cross-entropy loss is extensively used as the optimization criterion for HSI classification. The cross-entropy loss with softmax function can be written as

$$\mathcal{L}_{\text{softmax}} = -\sum_{i=1}^{m} \log \frac{\exp\left(W_{y_i}^T x_i + b_{y_i}\right)}{\sum_{j=1}^{C} \exp\left(W_j^T x_i + b_j\right)} \tag{12}$$

where m is the batch size, x_i is the ith feature correlating with the y_ith class, W_j is the jth column of the weight W in the fully connected layer, and b is the bias. To further improve feature discrimination and solve the challenge of high intraclass difference and low interclass discrepancy, center loss [49] is used to assist cross-entropy loss. Center loss can be formulated as

$$\mathcal{L}_{\text{center}} = \frac{1}{2}\sum_{i=1}^{m} ||x_i - c_{y_i}||_2^2 \tag{13}$$

where c_{y_i} is the center feature of the y_ith class. The function makes the feature of each sample close to the central feature of its category to decrease the intraclass distance. Consequently, the loss function for MSRN is formulated as

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{softmax}} + \lambda \mathcal{L}_{\text{center}} \\ &= -\sum_{i=1}^{m} \log \frac{\exp\left(W_{y_i}^T x_i + b_{y_i}\right)}{\sum_{j=1}^{C} \exp\left(W_j^T x_i + b_j\right)} + \frac{\lambda}{2}\sum_{i=1}^{m} ||x_i - c_{y_i}||_2^2\end{aligned} \tag{14}$$

where $\lambda \in (0, 1]$ balances the above two loss functions, and the value of λ is determined by experiments, as discussed in Section IV.

B. Multiscale Dense Block

Inspired by dense blocks, we improved the MSRB and designed the MSDB by combining the dual bypass structure and dense connections. The graphical illustration of MSDB is shown in Fig. 5. Conv1 and Conv3 are two layers of the first scale, and their convolution kernels have the same size to extract deep features at the same scale. Similarly, Conv2 and Conv4 belong to another scale. Different from MSRB, the second layer of convolution input not only has the output of the previous layer of convolutions (X_1, X_2), but also includes the input of MSDB (X_0). Inputs are concatenated together by the concatenation operation. After the bottleneck layer, the feature map with reduced dimensions is concatenated with the input X_0 as the output of the entire MSDB. The operations of MSDB can be written as

$$X_1 = W_{01} * g(X_0) \tag{15}$$
$$X_2 = W_{02} * g(X_0) \tag{16}$$
$$X_3 = [W_{03}, W_{13}, W_{23}] * g([X_0, X_1, X_2]) \tag{17}$$
$$X_4 = [W_{04}, W_{14}, W_{24}] * g([X_0, X_1, X_2]) \tag{18}$$
$$X_5 = [W_{35}, W_{45}] * g([X_3, X_4]) \tag{19}$$
$$X_6 = [X_0, X_5]. \tag{20}$$

Compared with MSRB, the top two feature extraction layers in MSDB can directly access to the input (X_0). Dense connection improves information flow and gradient flow. Meanwhile, the connections between bypasses of different scales enable the block to extract adaptive information at different scales.

We built a new spectral–spatial multiscale dense network (MSDN) on the basis of MSRN, as shown in Fig. 6. The first spectral down-sampling layer is used to extract initial spectral features and reduce spectral dimension. MSDN is divided into two MSDBs with the transition layers. The spectral MSDB reuses and refines spectral information through multiscale feature fusion and dense connections. Transition layers conduct spatial down-sampling and reshaping for spectral feature maps. Spatial MSDB utilizes spatial convolution kernels of different sizes to detect high-level spatial local information, which contributes to solve the problem that different land covers of HSI often have inconsistent size. After a 3-D average pooling layer and a fully connected layer, the vector $\hat{y}$ is devoted to calculate the loss between predictions and labels with (14) and optimize MSDN parameters by iterative training.

C. Multiscale Alternately Updated Clique Block

Previous studies [18], [19], [27] have shown that improving the information flow and parameter efficiency of the network can improve the network performance and reduce the training difficulty. Compared with residual connection,

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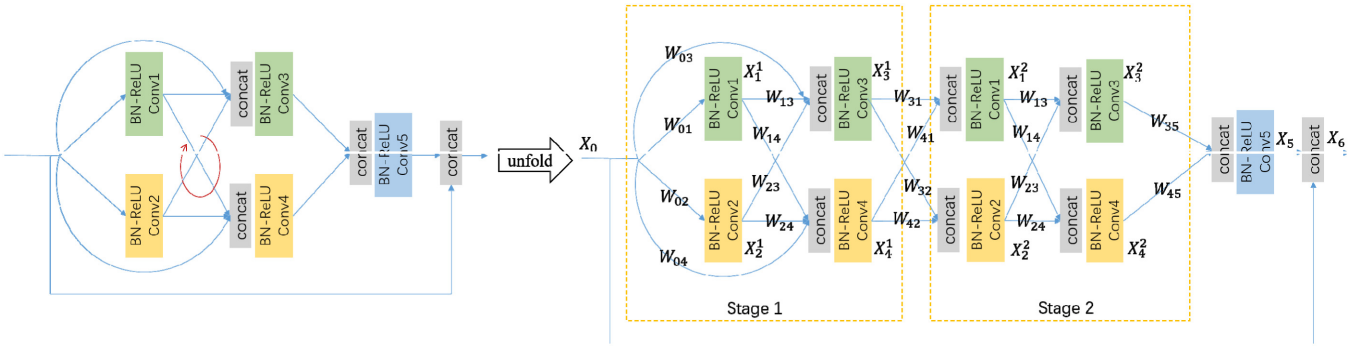


Fig. 7. Structure of MSCB.

the aggregation of features makes the gradient of each layer be directly accessed by the loss function. However, the number of parameters increases sharply with more dense layers being used. In order to further improve the information flow without increasing the dense layer, while alleviating the dependence of training more parameters on the number of training samples, we added a recurrent feedback structure on the basis of our MSDB and constructed multiscale alternately update clique block (MSCB). The structure of MSCB and its expansion are shown in Fig. 7. The recurrent feedback structure is set between the multiscale convolutional layers to maximize the information flow. Similar to clique blocks, the feature extraction process of MSCBs can be partitioned into different stages.

The first stage is to initialize top two layers in MSCBs, which explore the primary multiscale information. The formulaic expression of operations in Stage 1 is similar to MSDB

$$X_1^1 = W_{01} * g(X_0) \tag{21}$$
$$X_2^1 = W_{02} * g(X_0) \tag{22}$$
$$X_3^1 = [W_{03}, W_{13}, W_{23}] * g\left(\left[X_0, X_1^1, X_2^1\right]\right) \tag{23}$$
$$X_4^1 = [W_{04}, W_{14}, W_{24}] * g\left(\left[X_0, X_1^1, X_2^1\right]\right) \tag{24}$$

where X_i^1 denotes the output of Conv$i$ in Stage 1. At the first stage, outputs of all previous layers including other scales are concatenated as inputs of subsequent layers. The information flow of forward propagation is maximized. At the second stage, feedback connections are incorporated to refine features in different scales. We take advantage of the classic idea of alternate update, regard the multiscale information of the first stage as the input of the second stage, and reupdate the features. These multiscale layers construct a loop. The update process of the second stage can be formulated as

$$X_1^2 = [W_{31}, W_{41}] * g\left(\left[X_3^1, X_4^1\right]\right) \tag{25}$$
$$X_2^2 = [W_{32}, W_{42}] * g\left(\left[X_3^1, X_4^1\right]\right) \tag{26}$$
$$X_3^2 = [W_{13}, W_{23}] * g\left(\left[X_1^2, X_2^2\right]\right) \tag{27}$$
$$X_4^2 = [W_{14}, W_{24}] * g\left(\left[X_1^2, X_2^2\right]\right) \tag{28}$$

where X_i^2 denotes the output of Conv$i$ in Stage 2. MSCB enables bidirectional information flow between any two layers of multiscale layers. From Stage 2, multiscale layers begin updating alternately. In multiscale layers, all layers except current layer are concatenated as input layers, and their

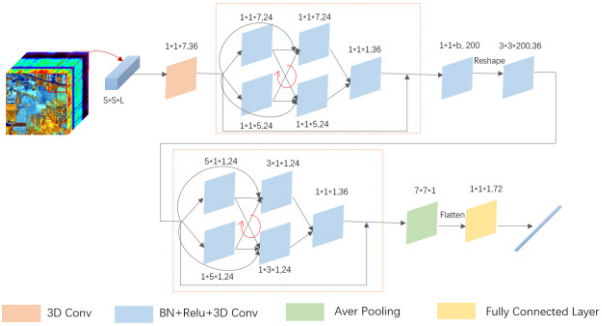


Fig. 8. Overview of spectral–spatial multiscale alternately updated clique network.

corresponding parameters are also concatenated. Multiscale feature maps and parameters are refined and reused. After the alternate update, the bottleneck layer is utilized to decrease the feature channel. The output of MSCB can be written as

$$X_5 = [W_{35}, W_{45}] * g\left(\left[X_3^2, X_4^2\right]\right) \tag{29}$$
$$X_6 = [X_0, X_5]. \tag{30}$$

Compared with MSDB, MSCB can implement more multiscale convolutional layers when increase of less parameters.

We replaced MSDBs with MSCBs to construct the MSCN. Fig. 8. displays the architecture of MSCN. It also has two MSCBs, which extract spectral and spatial multiscale information, respectively. It is worth noting that we improve the MSCB to MSCB-spa to decrease network parameters in spatial block. Sizes of convolution kernels in the same scale are consistent. In MSCB, both Conv1 and Conv3 are of size $a_1 \times a_1 \times b_1$, and Conv2 and Conv4 are of size $a_2 \times a_2 \times b_2$. Therefore, a convolution kernel has $a_i^2 b_i$ parameters, where i is 1 or 2. In [47], the consecutive 1 × n convolution and n × 1 convolution take the place of n × n convolution. This strategy saves computational consumption while ensuring performance. We apply spatial factorization into asymmetric convolutions to MSCB and design the MSCB-spa to specifically extract hyperspectral spatial features, where the convolution kernel sizes of Conv1 and Conv2 are $a_1 \times 1 \times b_1$ and $1 \times a_1 \times b_1$, respectively, and the convolution kernel sizes of Conv3 and Conv4 are $a_2 \times 1 \times b_2$ and $1 \times a_2 \times b_2$, respectively. The number of parameters in each convolution kernel is reduced to $a_i b_i$. This also reduces the possibility of overfitting.

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TABLE I
PARAMETER QUANTITIES OF THE MULTISCALE BLOCKS

Param	MSRB	MSDB	MSCB	
			MSCB	MSCB-spa
W_{01}	$k(a_1^2 b_1)$	$k(a_1^2 b_1)$	$k(a_1^2 b_1)$	$k(a_1 b_1)$
W_{02}	$k(a_2^2 b_2)$	$k(a_2^2 b_2)$	$k(a_2^2 b_2)$	$k(a_2 b_2)$
W_{03}	\	$k(a_1^2 b_1)$	$k(a_1^2 b_1)$	$k(a_1 b_1)$
W_{13}	$k(a_1^2 b_1)$	$k(a_1^2 b_1)$	$\boldsymbol{k(a_1^2 b_1)}$	$\boldsymbol{k(a_1 b_1)}$
W_{23}	$k(a_1^2 b_1)$	$k(a_1^2 b_1)$	$\boldsymbol{k(a_1^2 b_1)}$	$\boldsymbol{k(a_1 b_1)}$
W_{04}	\	$k(a_2^2 b_2)$	$k(a_2^2 b_2)$	$k(a_2 b_2)$
W_{14}	$k(a_2^2 b_2)$	$k(a_2^2 b_2)$	$\boldsymbol{k(a_2^2 b_2)}$	$\boldsymbol{k(a_2 b_2)}$
W_{24}	$k(a_2^2 b_2)$	$k(a_2^2 b_2)$	$\boldsymbol{k(a_2^2 b_2)}$	$\boldsymbol{k(a_2 b_2)}$
W_{31}	\	\	$k(a_1^2 b_1)$	$k(a_1 b_1)$
W_{41}	\	\	$k(a_1^2 b_1)$	$k(a_1 b_1)$
W_{42}	\	\	$k(a_2^2 b_2)$	$k(a_2 b_2)$
W_{32}	\	\	$k(a_2^2 b_2)$	$k(a_2 b_2)$
W_{35}	k_0	k_0	k_0	k_0
W_{45}	k_0	k_0	k_0	k_0
Total	$3k(a_1^2 b_1 + a_2^2 b_2) + 2k_0$	$4k(a_1^2 b_1 + a_2^2 b_2) + 2k_0$	$6k(a_1^2 b_1 + a_2^2 b_2) + 2k_0$	$6k(a_1 b_1 + a_2 b_2) + 2k_0$

Table I illustrates the number of parameters for each multiscale block, where k_0 and k represent the channel numbers of convolution kernels in the bottleneck layer, $(a_1 \times a_1 \times b_1)$ and $(a_2 \times a_2 \times b_2)$ denote two scales, and the reused parameters are bolded. The parameters of MSRB are less than MSDB and MSCB. MSDB has about 33% more parameters than MSRB due to dense connections. This is the price of improving information flow. The parameters of MSCB are about 1.5 times that of MSDB. However, through the unfolded structure of MSCB in Fig. 7, we can see that the multiscale layer of MSCB is twice that of MSDB. Although MSCB has more parameters, the efficiency of parameters and data flow within the network have been greatly improved. This is due to the feedback mechanism and the reuse of parameters.

IV. EXPERIMENTAL RESULTS

In this section, the validity of the presented multiscale networks including MSRN, MSDN, and MSCN is verified using three hyperspectral scenes including the Indian Pines (IP), the University of Pavia (UP), and Botswana (BS). Experiments have been implemented to compare several methods of HSI classification, including the contextual deep CNN (CDCNN) [41], the MSDNet [37], the spectral–spatial residual network (SSRN) [38], and the fast dense spectral–spatial convolution (FDSSC) framework [39]. CDCNN and MSDNet are the multiscale deep learning methods. The former jointly exploits local spatial–spectral interactions through a multiscale filter bank at the initial stage of the network. The latter extracts deep features in the horizontal dimension and the vertical dimension by the dense connectivity to realize the reconstruction of depth feature extraction and multiscale information fusion. Both of SSRN and FDSSC use continuous spectra and spatial blocks to extract discriminative information. The difference is that SSRN employs the residual architecture and FDSSC utilizes the dense structure. In order to survey the problem of high dimension with deficient samples, a small amount of samples is chosen at random for training, and the rest are used for testing and validation. The training set is devoted to optimize model parameters through backpropagation. The validation set does not participate in training, but is used to evaluate the temporary model of training phase through classification accuracies and losses. The weight parameters with the best verification accuracy are saved as the final model parameters. The testing set evaluates the classification ability of the trained model. Besides, each experiment is repeated ten times to prevent any deviation. The overall accuracy (OA), the average accuracy (AA), and the kappa coefficient are used as evaluative criteria. The experiments are implemented with the CPU of Intel E5-2650, the GPU of GTX2080Ti, and the memory of 64 GB.

TABLE II
SIXTEEN GROUND-TRUTH CLASSES IN IP AND THE LABELED NUMBER OF EACH CLASS

Class	Name	Samples
1	Alfalfa	54
2	Corn-no till	1434
3	Corn-min till	834
4	Corn	234
5	Grass-pasture	497
6	Grass-trees	747
7	Grass-pasture-mowed	26
8	Hay-windrowed	489
9	Oats	19
10	Soybean-no till	968
11	Soybean-min till	2468
12	Soybean-clean till	614
13	Wheat	212
14	Woods	1294
15	Buildings-grass-trees	380
16	Woods	95
Total		10366

A. Experimental Datasets

IP as the first dataset of our experiment was obtained by airborne visible infrared imaging spectrometer (AVIRIS) sensor in Northwestern Indiana. IP contains 145 × 145 pixels and 224 spectral reflectance bands with the wavelength range of 0.4–2.5 μm. We experiment 200 bands apart from noisy and water absorption bands. This dataset consists of 16 ground-truth categories as shown in Table II. We select the same amount of samples from per class to train. Since labeled samples of some classes are less than experimental settings, we only select nine classes which a total number of samples are greater than 400. The false color image and

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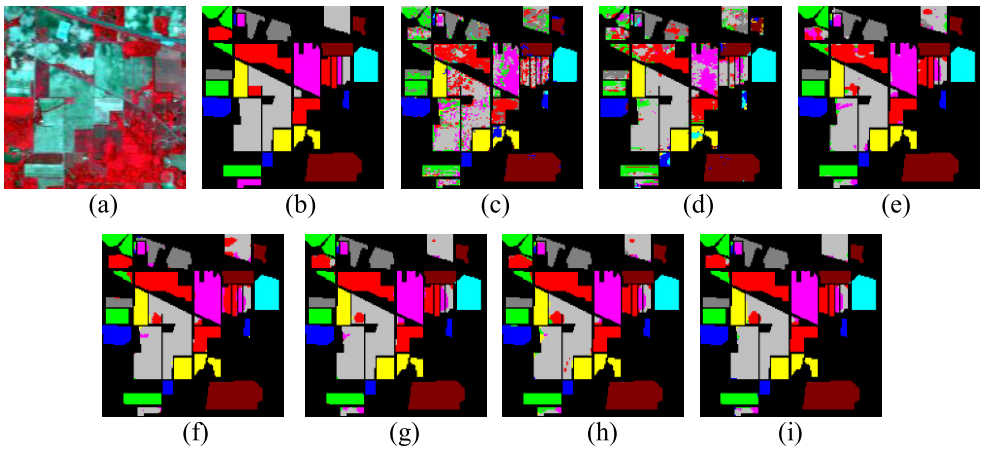


Fig. 9. For IP image. (a) RGB composite image of three bands. (b) Ground-truth map. (c) CDCNN. (d) MSDNet. (e) SSRN. (f) FDSSC. (g) MSRN. (h) MSDN. (i) MSCN.

TABLE III
NINE GROUND-TRUTH CLASSES IN UP AND THE LABELED NUMBER OF EACH CLASS

Class	Name	Samples
1	Asphalt	6631
2	Meadow	18649
3	Gravel	2099
4	Trees	3064
5	Metal sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadows	947
Total		42776

TABLE IV
FOURTEEN GROUND-TRUTH CLASSES IN BS AND THE LABELED NUMBER OF EACH CLASS

Class	Name	Samples
1	Water	270
2	Hippo grass	101
3	Floodplain grasses1	251
4	Floodplain grasses2	215
5	Reeds1	269
6	Riparian	269
7	Fierscar2	259
8	Island interior	203
9	Acacia woodlands	314
10	Acacia shrublands	248
11	Acacia grasslands	305
12	Short mopane	181
13	Mixed mopane	268
14	Exposed soils	95
Total		3248

ground-truth map of nine classes are indicated in Fig. 9(a) and (b), respectively.

The UP dataset, the second dataset, was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor in the Pavia, northern Italy. The size of UP is 610 × 340 pixels and 103 spectral bands. The ground truth image includes 9 classes as shown in Table III. The false color image and ground-truth map are indicated in Fig. 10(a) and (b), respectively.

BS dataset was captured by the NASA EO-1 satellite over the Okavango Delta in BS, which encompasses 1476 × 256 pixels and 145 spectral bands. The dataset was made up of 14 ground-truth types, which represent seasonal swamps, occasional swamps, and drier woodlands located in the distal portion of the Delta. Table IV indicates the numbers of each class. The false color image and ground-truth map are indicated in Fig. 11(a) and (b), respectively.

Due to lack of samples, the numbers of training samples and verification samples are set to a small range. In our proposed methods including MSRN, MSDN, and MSCN, the input spatial size is fixed at 9 × 9. We set the batch size and the number of epochs to 16 and 400, respectively. The learning rate is 0.0001 for IP dataset and 0.0003 for UP and BS datasets. We apply the combination of scales of the convolution kernel size of 1 × 1 × 5 and 1 × 1 × 7 for feature extraction in spectral dimension, and the combination of 3 and 5 spatial scales to acquire spatial features.

B. Experimental Results

For the IP dataset, 20 and 10 samples per class are chosen for training and validation, respectively. Table V demonstrates the detailed classification results of different networks for IP, and Fig. 9(c)–(i) displays the corresponding classification maps. In Table V, the best accuracy is shown in bold and our proposed multiscale networks achieve the superior results. In terms of OA, the proposed multiscale networks are obviously better and more stable than other methods, and MSCN is also higher than the other two multiscale methods MSRN and MSDN, reaching the highest accuracy of OA performance as 97.41%. The OA value produced by the MSDN is slightly lower than MSRN, but the classification accuracy for class 12 (Soybean-clean till) reaches the best result. It is worth mentioning that the classification accuracy of MSCN for class 8 (Hay-windrowed) reaches 100%. Two multiscale models compared offer poor results for class 2 (Corn-no till), 3 (Corn-min till), 10 (Soybean-no till), and 12 (Soybean-clean till).

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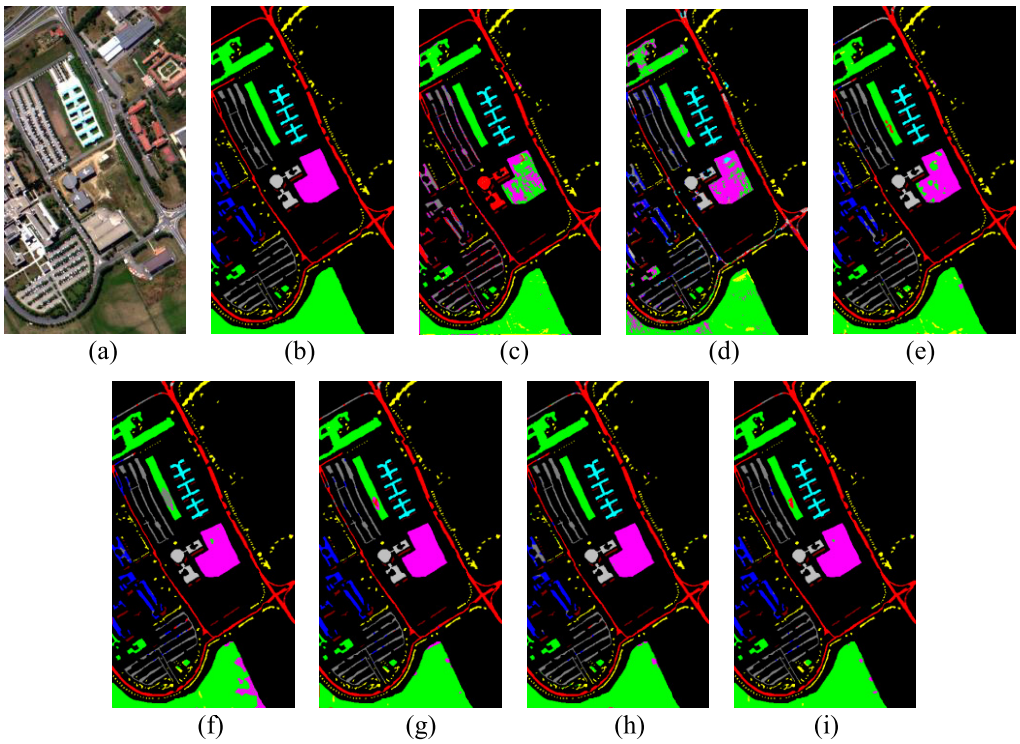


Fig. 10. For UP image. (a) RGB composite image of three bands. (b) Ground-truth map. (c) CDCNN. (d) MSDNet. (e)SSRN. (f)FDSSC. (g) MSRN. (h) MSDN. (i) MSCN.

TABLE V
CLASSIFICATION ACCURACY FOR IP USING DIFFERENT CLASSIFIERS

CLASS	COLOR	CDCNN [41]	MSDNet [37]	SSRN [38]	FDSSC [39]	MSRN	MSDN	MSCN
2		59.78 ± 11.86	74.00 ± 10.7	87.36 ± 9.78	93.0 ± 4.43	95.71 ± 1.9	95.56 ± 2.64	**96.4 ± 1.65**
3		53.24 ± 11.25	53.10 ± 6.87	96.81 ± 2.69	97.39 ± 2.55	98.29 ± 3.11	98.91 ± 0.74	**99.15 ± 0.62**
5		84.67 ± 18.58	92.18± 7.19	97.76 ± 1.73	97.4 ± 1.76	**97.99 ± 1.69**	97.88 ± 1.85	97.9 ± 1.96
6		91.4 ± 7.17	90.95± 2.92	**99.77 ± 0.21**	98.88 ± 0.94	99.75 ± 0.2	99.68 ± 0.29	99.6 ± 0.4
8		99.19 ± 0.69	97.70±2.02	99.9 ± 0.22	99.91 ± 0.15	99.98 ± 0.07	99.98 ± 0.07	**100.0 ± 0.0**
10		51.08 ± 20.62	62.68± 7.05	89.32 ± 4.27	91.06 ± 4.0	93.78 ± 1.93	93.45 ± 2.38	**94.82 ± 2.06**
11		63.71 ± 6.87	80.57 ± 4.88	85.14 ± 7.61	93.57 ± 4.09	95.35 ± 2.4	95.11 ± 2.06	**95.65 ± 1.78**
12		56.39 ± 19.62	52.78± 5.72	97.01 ± 1.34	94.78 ± 2.34	98.87 ± 1.36	**99.02 ± 1.49**	98.92 ± 1.41
14		91.04 ± 7.17	97.49 ± 0.70	99.14 ± 0.84	98.9 ± 1.31	99.68 ± 0.47	99.73 ± 0.25	**99.77 ± 0.2**
OA		69.22 ± 2.11	75.15 ± 2.60	92.17 ± 1.76	95.31 ± 1.19	97.04 ± 0.38	96.98 ± 0.61	**97.41 ± 0.54**
AA		72.28 ± 3.54	77.94 ± 2.24	94.69 ± 1.32	96.1 ± 0.75	97.71 ± 0.36	97.7 ± 0.4	**98.02 ± 0.35**
κ×100		63.95 ± 2.65	71.17 ± 2.76	90.83 ± 2.05	94.48 ± 1.39	96.52 ± 0.44	96.44 ± 0.71	**96.96 ± 0.63**

Combined with Fig. 9(b), it can be seen that these categories have smaller areas between other categories, and the boundary conditions are complicated. Complex boundaries and scattered regions usually increase the classification difficulty of categories. Through Table V, it can be demonstrated that the application of multiscale information interactions can help to solve this problem. The proposed multiscale networks show relatively good classification capabilities for these classes. For the purpose of better observation of the classification results, the classification maps of several networks for IP are displayed in Fig. 9(c)–(i). Comparing Fig. 9(c)–(i), it becomes obvious that the maps of MSRN, MSDN, and MSCN match the ground-truth map better. The misclassified pixels of other models are mostly scattered inside the classification areas, especially in the class 2 (Corn-no till) and 11 (Soybean-min till). It is consistent with the numerical results of Table V. At the same time, compared with the proposed multiscale networks, the classification maps of other networks are not smooth enough and the category boundaries are not clear. The classification map of MSCN can maintain the continuity within the category better than MSRN and MSDN, and the edges are clearer.

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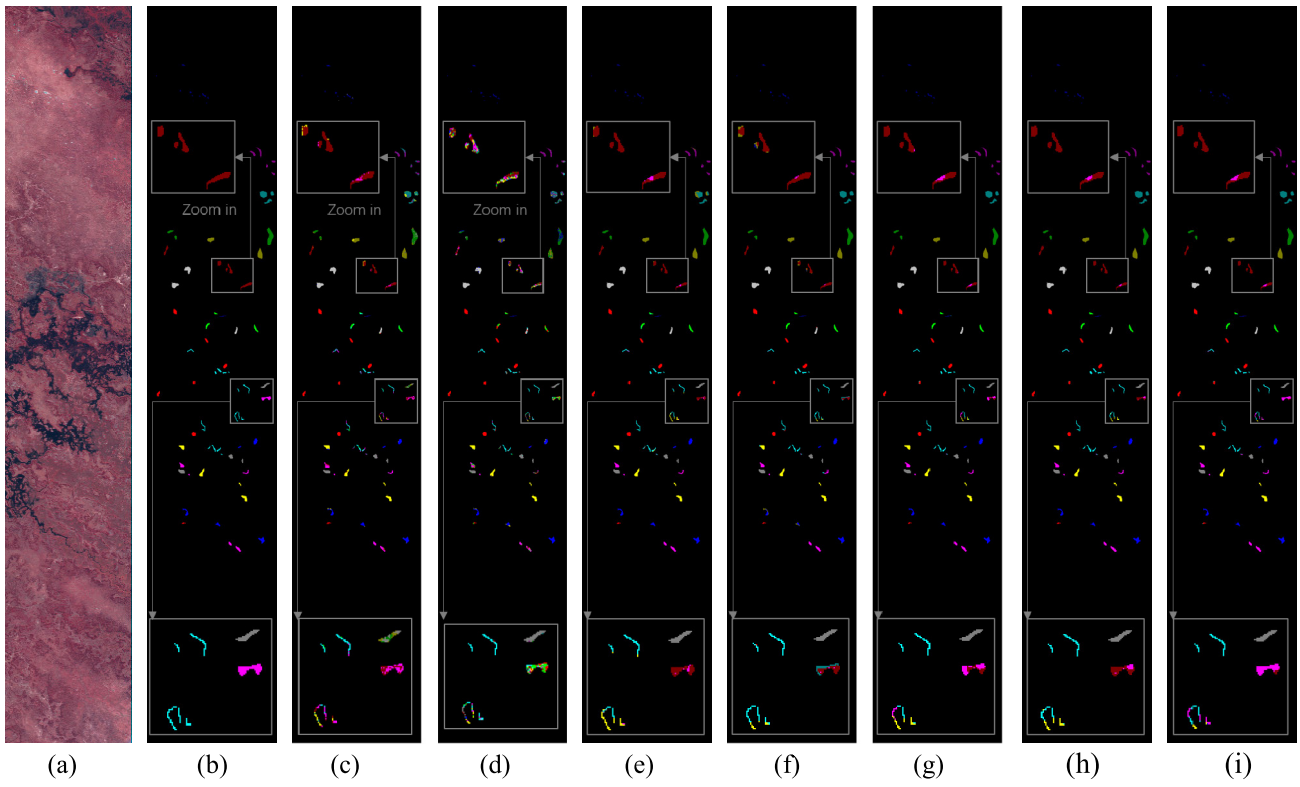


Fig. 11. For BS image. (a) RGB composite image of three bands. (b) Ground-truth map. (c) CDCNN. (d) MSDNet. (e) SSRN. (f) FDSSC. (g) MSRN. (h) MSDN. (i) MSCN.

TABLE VI
CLASSIFICATION ACCURACY FOR UP USING DIFFERENT CLASSIFIERS

CLASS	COLOR	CDCNN [41]	MSDNet [37]	SSRN [38]	FDSSC [39]	MSRN	MSDN	MSCN
1		80.63 ± 8.01	90.22 ± 5.31	93.32 ± 4.08	94.96 ± 4.66	96.47 ± 4.16	95.73 ± 3.23	**96.86 ± 2.52**
2		87.14 ± 6.59	87.09 ± 1.59	93.3 ± 3.8	94.06 ± 3.24	95.99 ± 2.65	97.28 ± 2.41	**98.72 ± 1.28**
3		65.13 ± 23.24	45.00 ± 6.47	91.01 ± 4.98	91.61 ± 4.29	**95.89 ± 4.0**	92.22 ± 5.23	88.91 ± 11.61
4		90.6 ± 6.27	94.50 ± 2.32	95.54 ± 2.67	96.2 ± 3.34	**96.9 ± 3.62**	95.51 ± 1.87	95.83 ± 3.35
5		99.92 ± 0.16	89.31 ± 2.79	**99.98 ± 0.05**	99.98 ± 0.05	99.97 ± 0.09	99.96 ± 0.06	99.86 ± 0.22
6		58.12 ± 10.36	50.09 ± 10.35	95.29 ± 3.92	99.43 ± 0.94	**99.76 ± 0.37**	98.97 ± 1.24	98.97 ± 0.98
7		76.97 ± 25.64	55.73 ± 8.90	99.33 ± 0.75	99.75 ± 0.53	**99.94 ± 0.1**	99.72 ± 0.33	99.82 ± 0.26
8		69.85 ± 13.89	69.59 ± 10.19	88.36 ± 9.68	95.71 ± 2.78	92.81 ± 5.68	95.28 ± 2.71	**95.6 ± 4.24**
9		97.48 ± 3.86	83.76 ± 11.58	98.29 ± 4.92	99.69 ± 0.44	**99.7 ± 0.35**	99.52 ± 0.56	99.43 ± 0.61
OA		80.72 ± 2.53	76.11 ± 3.67	93.66 ± 2.37	95.49 ± 1.64	96.62 ± 1.58	96.9 ± 1.29	**97.59 ± 0.93**
AA		80.65 ± 4.81	73.92 ± 1.62	94.93 ± 1.48	96.82 ± 0.74	**97.49 ± 1.01**	97.13 ± 0.7	97.11 ± 1.25
κ×100		74.6 ± 3.12	68.79 ± 4.08	91.7 ± 3.05	94.09 ± 2.12	95.56 ± 2.05	95.91 ± 1.69	**96.81 ± 1.23**

In the UP dataset, 20 and 10 samples per class are chosen for training and validation, respectively. Table VI and Fig. 10 report the numerical and graphical results, respectively. Through the smallest OA standard deviation, it can be found that the results of MSCN are more stable. From the accuracy of a single category, MSCN has the strongest ability to classify classes 1 (Asphalt), 2 (Meadows), and 8 (Self-Blocking Bricks). MSDN achieves the close OA to MSRN. However, MSRN has stronger performance classification of classes 3 (Gravel), 4 (Trees), 6 (Bare Soil), 7 (Bitumen), and 9 (Shadows). The highest AA of MSRN indicates its advantage in dealing with irregular and small objects. As shown in Fig. 10, the classification maps of CDCNN and MSDNet have quite a bit misclassification regions within classes 6 (Bare soil) and 7 (Bitumen). It shows that these two methods are difficult to extract more discriminative features in the case of limited samples. The category boundaries of FDSSC are not clear. In general, the classification maps of our proposed networks

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TABLE VII
CLASSIFICATION ACCURACY FOR BS USING DIFFERENT CLASSIFIERS

CLASS	COLOR	CDCNN [41]	MSDNet [37]	SSRN [38]	FDSSC [39]	MSRN	MSDN	MSCN
1		99.23 ± 0.97	68.90 ± 5.34	**100.0 ± 0.0**	99.38 ± 0.78	98.96 ± 1.32	99.85 ± 0.26	99.3 ± 1.07
2		86.62 ± 9.06	41.03 ± 5.94	98.73 ± 2.59	99.68 ± 0.97	**100.0 ± 0.0**	**100.0 ± 0.0**	99.79 ± 0.64
3		85.2 ± 14.6	64.52 ± 16.82	98.16 ± 3.04	95.24 ± 4.96	98.92 ± 2.59	**99.83 ± 0.38**	99.75 ± 0.76
4		70.85 ± 19.78	61.83 ± 7.92	98.93 ± 2.77	98.4 ± 2.14	**100.0 ± 0.0**	**100.0 ± 0.0**	99.8 ± 0.59
5		51.22 ± 16.6	79.53 ± 5.87	87.12 ± 9.57	84.72 ± 10.73	88.17 ± 7.37	**89.71 ± 12.55**	88.86 ± 7.12
6		67.35 ± 12.14	57.83 ± 4.70	80.22 ± 11.66	84.69 ± 11.14	91.13 ± 5.89	89.29 ± 7.85	**93.14 ± 3.95**
7		92.17 ± 8.85	96.29 ± 2.63	99.27 ± 1.83	99.67 ± 0.75	**100.0 ± 0.0**	**100.0 ± 0.0**	**100.0 ± 0.0**
8		72.41 ± 12.0	69.67 ± 14.35	94.32 ± 9.46	98.54 ± 2.56	96.85 ± 8.95	**99.69 ± 0.63**	96.74 ± 8.95
9		67.8 ± 12.78	65.96 ± 11.23	87.92 ± 10.64	96.79 ± 3.46	**97.49 ± 3.34**	95.97 ± 5.47	95.61 ± 6.59
10		72.52 ± 19.73	57.18 ± 11.12	96.12 ± 6.7	98.99 ± 1.64	96.18 ± 11.47	**99.79 ± 0.63**	98.62 ± 4.14
11		67.99 ± 23.12	67.37 ± 9.72	84.84 ± 13.31	87.7 ± 11.88	91.52 ± 10.61	87.96 ± 11.37	**92.29 ± 11.16**
12		91.26 ± 9.11	67.22 ± 13.58	98.6 ± 1.45	97.52 ± 4.27	99.76 ± 0.71	99.88 ± 0.24	**99.94 ± 0.18**
13		68.89 ± 23.55	65.73 ± 13.28	94.1 ± 11.86	98.6 ± 3.82	99.77 ± 0.69	99.73 ± 0.56	**100.0 ± 0.0**
14		89.35 ± 9.25	54.85 ± 13.68	94.55 ± 6.91	93.09 ± 7.21	91.83 ± 7.78	94.33 ± 6.56	**96.36 ± 5.01**
OA		75.7 ± 6.89	64.36 ± 3.93	93.0 ± 2.38	94.81 ± 0.75	96.31 ± 1.39	96.47 ± 1.32	**96.83 ± 1.41**
AA		77.35 ± 6.29	65.57 ± 3.44	93.78 ± 2.03	95.22 ± 0.65	96.47 ± 1.26	96.86 ± 1.13	**97.16 ± 1.21**
κ×100		73.72 ± 7.42	61.56 ± 4.23	92.41 ± 2.57	94.37 ± 0.81	96.0 ± 1.5	96.18 ± 1.43	**96.57 ± 1.52**

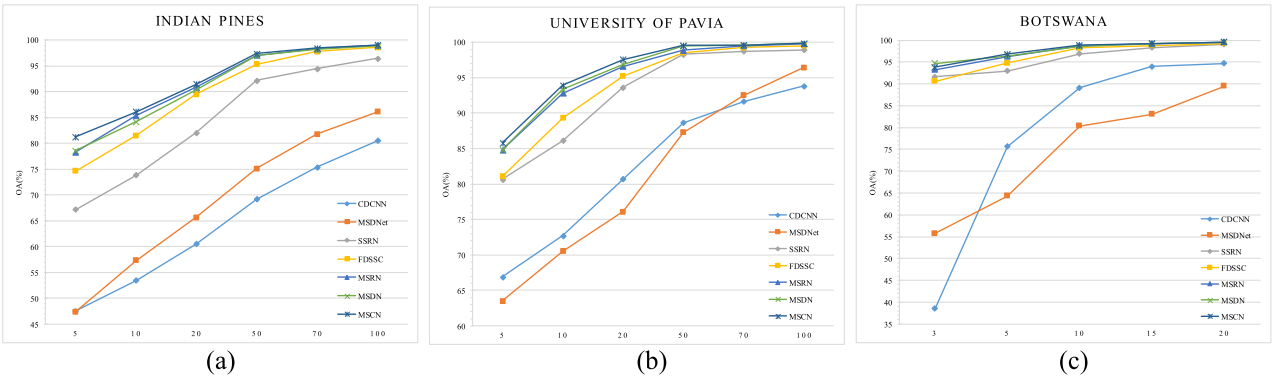


Fig. 12. OA of different methods for different numbers of training samples in three HSI datasets. (a) IP. (b) UP. (c) BS.

are smoother and more consistent with the ground-truth map in detail.

For the BS dataset, 5 samples per class are chosen for training and validation, respectively. The classification results with above approaches in the BS dataset are displayed in Table VII, in which the best class-specific accuracy is highlighted in bold, and classification maps are presented in Fig. 11(c)–(i). We zoomed-in view two regions that are easier to be misclassified in classification maps to improve the visual effect. As shown in Table VII, the proposed MSCN reaches the highest accuracy among all the other classifiers in items of OA, AA, and kappa coefficient. SSRN achieves the best result for class 1 (Water). However, in other categories and OA, SSRN and FDSSC are still inferior to the proposed multiscale methods. The OA of MSDNet is the most unsatisfactory, while the classification performance of CDCNN is the most unstable. Especially for category 7 (Fierscar2), the three proposed networks have reached 100% accuracy. In addition, MSDN and MSRN are 100% accurate in classification of classes 2 (Hippo grass) and 4 (Floodplain grasses2). MSCN achieves the 100% accuracy of class 13 (Mixed mopane). In this dataset, the OAs of three networks are not much different, but MSCN performs better than MSDN and MSRN. The size of the BS dataset is much larger than UP and IP, but the labeled samples are few and scattered. The classification performance on the BS datasets illustrates the excellent adaptability of the proposed multiscale method to different types of datasets.

Deep neural networks demand a great amount of training samples. To verify the robustness and stability of our proposed multiscale networks, we investigated the impact of different numbers of training samples on the network classification ability. Fig. 12(a)–(c) demonstrates the results

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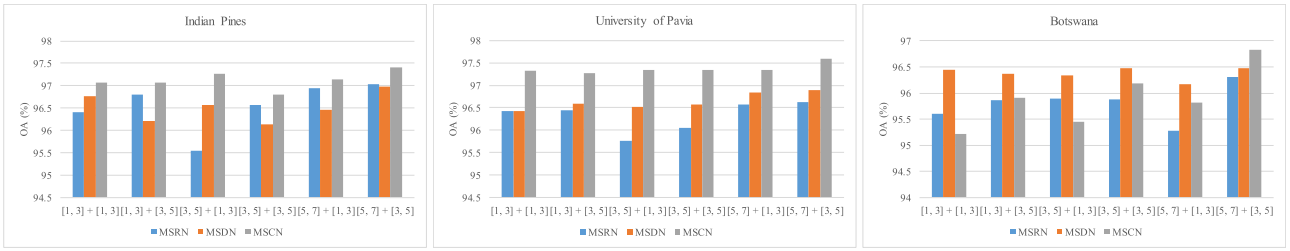


Fig. 13. OA of the three multiscale networks with different spatial sizes. (a) IP. (b) UP. (c) BS.

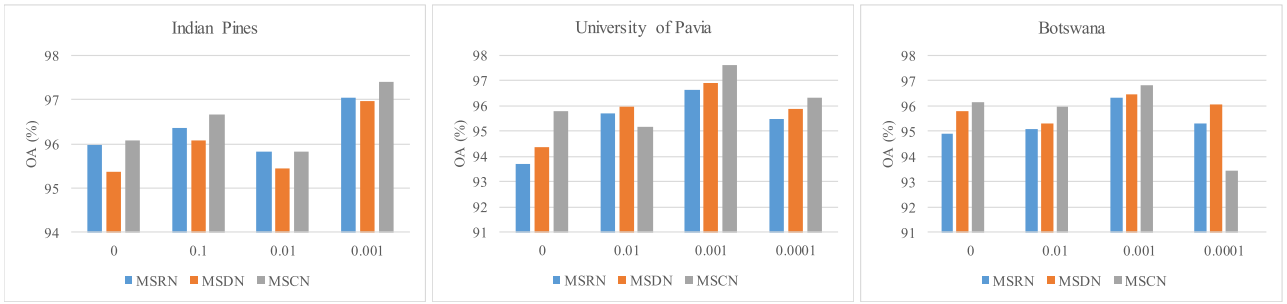


Fig. 14. OA of the three multiscale networks with different scale combinations. (a) IP. (b) UP. (c) BS.

for the three datasets. It can be found that whatever the number of training samples per class, the classification results of our proposed MSRN, MSDN, and MSCN in the three datasets are significantly better than others, especially under circumstance of fewer training samples. Although CDCNN and MSDNet also apply multiscale information, they are most sensitive to changes in the number of training samples. When training sample is less, it is difficult to obtain credible feature representation capabilities. When training samples of each class are greater than 50, the classification abilities of the proposed multiscale networks on the IP dataset have little difference. The classification performances of MSDN on the UP and BS datasets are better than MSRN, and the opposite on the IP dataset. MSCN always is superior to MSRN and MSDN, in addition to 5 training samples of the BS dataset. MSCN is more adaptable to different datasets.

The number of parameters and running time are important indicators for evaluating neural network performance. We take the BS as an example to display the training and testing time and the number of parameters of the above methods in Table VIII. Since the epoch is set to 400, our methods obviously spend more time than SSRN or FDSSC. Thanks to MSCB-spa, MSCN has fewer parameters than MSDN. However, due to repeated use of parameters, computing time has not been reduced. MSRN utilizes fewer parameters and obtains better results than FDSSC. MSDNet integrates information of different scales from coarse to fine levels, but also introduces a large number of parameters, which not only requires the longest training time, but also makes it difficult to achieve the ideal results in the case of small number of training samples. It indicates that improving parameter efficiency and extracting efficient information can effectively alleviate the network's dependence on training samples.

TABLE VIII
RUNNING TIMES AND THE NUMBER OF PARAMETERS FOR DIFFERENT METHODS FOR BS

Methods	Training Time (s)	Testing Time (s)	Parameters (M)
CDCNN	27.62	0.52	0.82
MSDNet	379.91	10.41	2.58
SSRN	52.37	1.57	0.28
FDSSC	89.82	1.64	0.90
MSRN	174.92	2.06	0.67
MSDN	251.49	2.06	1.21
MSCN	316.62	3.56	1.16

C. Influences of Scale Combinations, Loss Weight, and Spatial Size

As an important element of feature extraction in our proposed framework, the scale combinations of the multiscale structures directly affect the classification performance. We experiment with different scale combinations in Fig. 13. The two items before and after "+" represent the scale combinations of the spectral block and the spatial block, respectively. It can be seen that the MSCN in the UP dataset is insensitive to scales. In general, the scale combination of "[5,7] + [3,5]" is the best choice.

Fig. 14 shows the impact of the loss weight λ. Three multiscale networks are all sensitive to changes in loss weights. We choose $\lambda = 0.001$, because it can obtain the best results on all three datasets.

The spatial size of the input patch, as another key factor, has a great influence on classification accuracy. We experiment with different spatial sizes, as shown in Fig.15. According to the results on three datasets, it can be observed that

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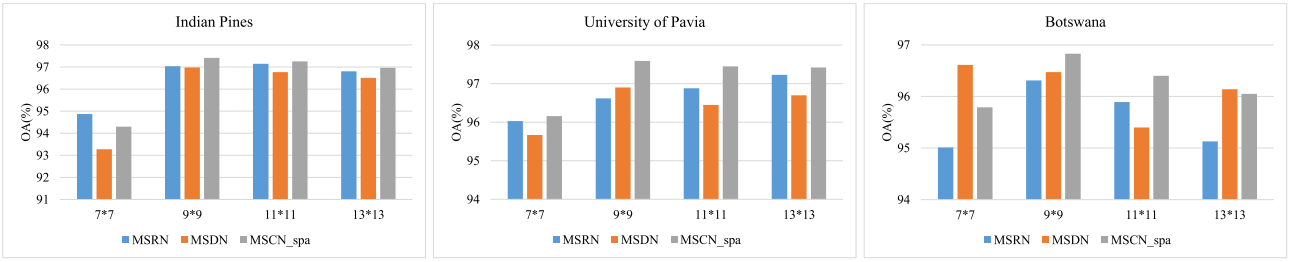


Fig. 15. OA of the three multiscale networks with different loss weights. (a) IP. (b) UP. (c) BS.

TABLE IX
EFFECTIVENESS OF CROSS FUSION AND RECURRENT FEEDBACK MECHANISMS

	Cross Fusion	Feats-I	Feats-II	IP	UP	BS
MSCN (only with MSCBs)	√	√	×	**96.57**	96.48	95.95
	×	×	√	96.09	96.47	95.61
	√	×	√	95.94	**96.99**	**96.06**
MSCN (with MSCB-spa)	√	√	×	96.74	97.18	95.00
	×	×	√	96.74	96.69	95.94
	√	×	√	**97.41**	**97.59**	**96.83**

when the patch size is set to 7×7, the accuracies of the multiscale networks on three datasets are usually lower than those from using other spatial sizes. When the patch size is greater than 9×9, the accuracies are not improved remarkably, but the calculation cost is increased significantly. Therefore, considering the tradeoff between performance and computational overhead, we choose 9×9 as the patch size.

D. Ablation Study for Cross Fusion and Recurrent Feedback Mechanisms

In this section, some ablation experiments are conducted to discuss three issues: 1) the effectiveness of MSCB-spa; 2) the effectiveness of the cross fusion mechanism on different scales; and 3) whether the recurrent feedback mechanism can refine features effectively.

Based on the results in Table IX, using MSCB-spa to replace the space MSCB can achieve better classification performance under the same conditions. In most cases, integrating information only at the end of the multiscale block reduces the representation ability. The information fusion inside the multiscale block provides more local details and enriches contextual information. When the Stage-1 features (Feats-I) of MSCB are delivered to the bottleneck layer at the end of the block, the results obtained are not as good as using the Stage-2 features (Feats-II). It demonstrates that feature rerefinement based on parameter reuse can indeed help the networks extract more abstract features and further improve the performance of the networks.

V. CONCLUSION

This article proposed three multiscale networks based on multiscale residual, dense, and clique structures to utilize convolution kernels of different sizes and adaptively capture the contexture information at multiple scales. MSRN, MSDN, and MSCN exploit corresponding multiscale blocks to capture spectral and spatial multiscale information successively in a sequential connection manner. Experiments on three representative hyperspectral datasets demonstrated that the fusion and application of spectral–spatial features at different scales make our methods superior to other deep learning approaches and achieve higher classification accuracy, especially under the situation of fewer training samples.

Compared with MSRN, dense connection of MSDN improves information flow and also allows its parameters to increase rapidly. MSDN offers more powerful classification capabilities in irregular regions. Compared with MSDN, the recurrent feedback mechanism and the spatial factorization into asymmetric convolutions of MSCN maximize the information flow and parameter efficiency of the network. Reuse of high-level feature and parameters allow MSCN to achieve higher results with fewer parameters and reduce the dependence of parameters on training samples. Thus, MSCN provides the best classification results. However, MSCN is the most time-consuming in parameter optimization. Reducing network training time will be the direction of our future work.

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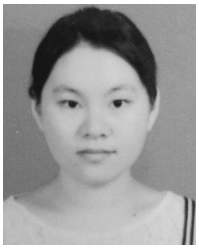
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